

Research Article

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Fiber-reinforced concrete for complex engineering conditions: material properties, nonlinear response and sustainable applications in spaceflight

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Abstract: As space missions evolve toward longer durations and deeper missions, structural materials face unprecedented challenges in extreme extraterrestrial environments, such as extreme temperature swings, intense radiation, cyclic loading, and crack propagation. These conditions place higher demands on the ductility, energy dissipation, and adaptability of traditional brittle concrete, prompting the research community to seek material systems with enhanced nonlinear response capabilities and sustainability. As a brittle material, ordinary concrete lacks the toughness and adaptability required to sustain such nonlinear behaviors. Fiber-reinforced concrete (FRC) has gained increasing attention for its ability to enhance material ductility, energy dissipation, and structural resilience under nonlinear conditions. This review summarizes recent research on FRC, focusing on its mechanical properties, nonlinear response mechanisms, and potential for sustainable construction. The inclusion of natural, synthetic, or hybrid fibers significantly improves compressive strength, flexural performance, and crack control. Moreover, the use of recycled fine aggregates from construction and demolition waste further promotes environmental sustainability. The combined effects of fiber type, volume fraction, and distribution on nonlinear load-bearing capacity are critically analyzed. Finally, this article proposes that future research should focus on the performance evolution and adaptive design strategies of FRCs in complex nonlinear service environments (including high strain rates and low gravity in

aerospace), providing theoretical support and material pathways for building the next generation of durable, lightweight, and sustainable aerospace infrastructure. FRC presents a promising solution for developing durable, adaptive, and sustainable infrastructure systems in nonlinear engineering environments.

Keywords: fiber-reinforced concrete; mechanical properties; nonlinear response

1 Introduction

Concrete plays an essential role in the construction of civil infrastructure and is the most widely used building material globally, largely due to its cost-effectiveness, longevity, and ease of application [1]. Since 2015, the concrete industry has seen a global demand for natural aggregates reaching 48.3 billion metric tons annually, with an approximate annual growth rate of 5 %. It can be predicted that concrete will still be an important engineering structural material for a long time to come. The utilization of natural resources in the construction of built environments has similarly triggered widespread apprehension within the context of sustainable development.

Concrete is primarily composed of cement, which acts as the binding agent, aggregates that provide structural support, and water, which is essential for the chemical reaction that hardens the mixture. Approximately 70–80 % of concrete's total volume is constituted by aggregates. Therefore, the attributes and qualities of aggregates directly affect the overall efficacy and functionality of concrete, influencing factors such as workability, strength, and durability.

With the increasing worldwide need for environmentally conscious progress, infrastructure is constantly being improved to meet the needs of a growing population and changing demands. The contemporary way of life has led to an increase in waste production and a significant rise in construction and demolition (C&D) waste, which poses disposal challenges due to advancements in technology. For

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example, China is known as the world's largest producer of construction and demolition waste, with an estimated annual output ranging from 18 to 35 billion tons. During 2018, American construction and demolition activities produced a substantial 600 million metric tons of debris. Meanwhile, Britain creates around 1.2 billion tonnes of C&D waste each year, representing more than three-fifths of the country's overall waste output [2]. Thus, recycling and reusing construction solid waste are crucial strategies for minimizing resource consumption and aiding in environmental conservation. This represents a significant advancement in eco-friendly building practices, addressing a widespread challenge effectively. Waste materials can be utilized as filler, fibers or substitutes for fine aggregate in concrete.

Recycled aggregate concrete (RAC) enhances the durability and lifespan of concrete, reduces specific energy usage, and incorporates recycled aggregates to lessen the environmental impact of these processes in order to fulfill these requirements [3, 4]. However, it also presents significant performance challenges. Concrete produced using C&D waste exhibits greater porosity, which leads to increased water absorption and reduced strength compared to concrete made with natural aggregates. This elevated water absorption also contributes to a weaker structural integrity.

To this end, scholars have proposed various methods for treating recycled aggregate, including acid leaching, carbonation, and nanosilica treatment, among others. However, these methods are rarely applied in actual engineering projects. Therefore, it is crucial to utilize recycled aggregate concrete in a rational manner. Extensive research has revealed that incorporating fibers, which serve as a bridging mechanism, can dramatically enhance the interfacial properties of reinforced concrete [4]. The enhancement of mechanical properties represents a notable benefit, as it can result in concrete that exhibits improved strength, deformation resistance, toughness, and durability. By incorporating fibers into RAC, a viable solution may be provided to overcome its inherent strength limitations.

Synthetic materials are challenged by environmental conditions (radiation, microgravity, extreme temperature variations, cyclic extraterrestrial conditions), and their mechanical properties exert nonlinear effects [5]. This renders traditional cointegration analysis methods for material-behavioral-mechanical correlations in structures ineffective. To overcome these limitations and address the time-consuming nature of concrete mix design experiments, numerous studies have developed predictive models under nonlinear conditions. For example, during the backfill reinforcement process, calcium sand particles will undergo cyclic deformation under load. The influence law of cyclic stress ratio and loading frequency on the elastic modulus

of soil needs to be nonlinear fitting, and the cumulative plastic deformation of different deformation states of materials is predicted. Voronoi-RBSM demonstrates significant advantages through its use of mesoscale numerical simulations to predict concrete's macroscopic mechanical properties, particularly its complex nonlinear behaviors. In the aerospace infrastructure, to resist the large temperature difference harsh environment, the model can generate digital aggregates with high accuracy that closely match real-world counterparts, while embedding zero-thickness bonding units at the fiber reinforced aggregate-cement mortar interface to precisely characterize the mechanical characteristics of the interfacial transition zone [6].

In the study of nonlinear concrete materials, researchers have meticulously established a wealth of full stress-strain curve equations derived from experimental data. These equations prove invaluable for analyzing and comparing complex stress and strain distribution patterns, enabling insightful qualitative discussions on how boundary conditions shape these distributions, and offering crucial references for designing metal fiber materials. For instance, Zhang and Zhang (2002) developed a novel viscoelastic theory and rigorously validated it using modified polypropylene materials. When designing materials destined for aviation, aerospace, and defense structures, it becomes imperative to account meticulously for their thermal behavior and assess their capacity to endure thermal cycling loads involving significant temperature fluctuations [7]. Rolfes pioneered a thermal stacking theory seamlessly integrated with finite element methods; subsequent modifications and advancements building upon this approach have furnished effective solutions for the intricate thermal analysis and design of composite materials [8]. Addressing structural nonlinear instability problems, the material-based equilibrium state parameter ϕ_t was innovatively introduced. This parameter leverages the derivative ratio law to tackle the formidable nonlinear aging challenges inherent in buckling and fracture phenomena, though deeper exploration in this domain demands participation from significantly more extensive experimental data [9].

The above theoretical research is based on laboratory measured data, however, conventional approaches like manual inspections have precision and efficacy issues due to human error and environmental impacts, AI-driven structural health monitoring (AI-SHM) employs machine learning and deep learning techniques to overcome the above issues [10].

In the study of mechanical properties of fiber-reinforced concrete, considerable research has focused on examining the impact of combining synthetic fibers with natural fibers on concrete characteristics. Shaikh et al. (2023) investigated the properties and compressive strength of RAC through the

incorporation of steel fibers [11]. Wang et al. (2023) reveals that the compressive strength rises initially and then diminishes with the augmentation of fiber volume and aspect ratio, and diminishes with the augmentation of RA substitution rate [12]. Fu et al. (2024) examined the mechanical properties of RAC by introducing waste glass fibers [13]. Ahmed and colleagues (2023) performed a comprehensive evaluation of freeze-thaw durability and mechanical characteristics in RAC through the integration of polypropylene, basalt, and glass fibers [14]. Separate research by Rattanachu et al. (2022) examined the effectiveness of rice husk ash as a binding component within recycled concrete mixtures [15]. In parallel investigations, Yan's research team (2023) analyzed the dynamic performance characteristics of fiber-reinforced recycled concrete using flax-based reinforcements [16]. Nili and Zhao et al. (2023) conducted a comprehensive study on fiber reinforced concrete, focusing on enhancing its mechanical properties and durability, as well as developing performance prediction methods and mix design techniques. They compared the reinforcing effects of various fiber-reinforced concretes and selected the optimal fibers at different scales. The study aimed to provide valuable insights into improving the overall performance of fiber reinforced concrete [17]. Mohana et al. (2024) studied the application of natural fibers in geopolymer concrete structures, utilizing a hybrid of natural and synthetic fibers [18]. Gulzar et al.'s 2023 research demonstrated that incorporating jute fibers at 0.5 % volumetric concentration enhanced concrete's bending resistance, showing a 17–21 % strength increase relative to standard mixtures. The study specifically measured performance under flexural stress conditions [19]. Mastali et al. (2019) investigated the impact strength of concrete reinforced with recycled tire steel fibers (at 0.5 % and 1.0 % dosages) [20]. Choi and Yuan (2020) discovered that incorporating polypropylene fiber (PPF) into normal vibrated concrete enhanced the STS by 20–50 % [21].

In the study of synthetic fiber nonlinear model, Gong and Li (2024) introduced a nonlinear bond slip model to simulate the synergy behaviour of reinforced steel bars and concrete to describe the weakening process of interface spring between adjacent blocks [6]. Zhao et al. (2024) proposed a new idea of concrete meso modelling based on Voronoi mesh topology and rigid body spring model (RBSM), which predict the meso mechanical response of concrete members in the loading process more efficiently and accurately [22]. Lim al. (2021) developed Faster R-CNN which combined with vision and infrared thermography to captured the details about the corrosion-induced damage of bridges on the surface and also sub-surface damage [23].

Fiber-reinforced cement composites typically diminish adverse ecological impacts while lowering resource

utilization. The integration of fibrous materials within RAC signifies substantial progress in building material technology, delivering superior longevity, adaptability, and operational characteristics relative to traditional cementitious mixtures. The incorporation of diverse fiber materials, including metallic, organic, polymer-based, and silica-based reinforcements, into cementitious composites markedly enhances their capacity to withstand tension, improves deformation tolerance, and fortifies protective characteristics across multiple dimensional levels. These features are crucial for structural integrity and load-bearing capacity.

This paper focused on reviewing the mechanical properties of natural fibers and synthetic fibers which are fiber-reinforced RAC composites. Consequently, the review's findings offer valuable insights into the potential advantages and constraints of incorporating these recycled materials and fibers into new concrete production. Furthermore, this comprehensive comparative analysis can act as a crucial reference for future research, aiding in the creation of more sustainable and eco-friendly concrete solutions.

2 Fibers

Fibers are categorized into two types: natural and synthetic. These materials are categorized into two primary forms: either short/discontinuous strands (ranging from 5 to 50 mm) or uninterrupted filaments, as referenced in study [24].

2.1 Natural fibers

Natural fibers exhibit decreased mass-to-volume ratios, economical production costs, improved shock absorption capabilities, and exceptional pliability. Their distinguishing features encompass minimal density measurements, diminished machinery wear, reduced occupational risks, improved manufacturing compatibility, lower atmospheric pollutant outputs, and complete sustainability through recyclability and carbon balance [25]. Natural fibers like straw, hemp, and kenaf have been employed in construction for multiple structural applications, including reinforcement for earthen walls, ropes for constructing bridges, roofing materials, and other architectural uses.

Wang et al. (2023) research demonstrates that integrating certain plant-derived fibers, including jute, coconut, and sisal, substantially enhances the structural integrity of cementitious composite materials [26]. Incorporating chopped natural fibers into concrete significantly mitigates shrinkage while enhancing both the toughness and strength of the material. The analysis indicates that multiple organic

Table 1: Advantages and disadvantages of natural fibers.

Advantages	Disadvantages
Low density	Low strength
Exceptional wear resistance	Highly hygroscopic
High toughness	Flammable
Outstanding strength relative to weight	Restricted thermal processing range (200–230 °C)
Biodegradable	Prone to fungal and microbial attack
Shrinkage resistance	Low durability
Bridging in cracks	
Insulation	

fibers derived from plants demonstrate significant potential for enhancing the structural integrity of concrete composites [27]. For example, adding 0.5 % jute fiber by volume to concrete has been shown to increase flexural strength by 17–21 % compared to standard concrete [28]. The advantages and disadvantages of natural fibers are summarized in Table 1.

Nevertheless, integrating botanical fibers into cementitious mixtures introduces specific difficulties. The interfacial bonding characteristics between these natural reinforcements and the cement matrix prove crucial, since they substantially influence the structural integrity of the resulting material. Furthermore, questions persist regarding the extended service life of plant fiber composites, especially in scenarios demanding high tensile performance and prolonged structural resilience. Taking bamboo fibers as an example, impurities such as hemicellulose, pectin, partial lignin, and waxy substances can interfere with the cement hydration process, preventing the formation of adequate bonding strength. Only modified bamboo fibers exhibit increased specific surface area and enhanced roughness, which facilitates their expansion and mineralization within the cement matrix. This bridging effect significantly improves the material's overall

mechanical properties. Figure 1 shows the FE-SEM images of bamboo fibers before and after modification.

2.2 Synthetic fibers

Synthetic fibers can be organic based (carbon and aramid) or inorganic based (glass and basalt). The integration of synthetic fibers into concrete has been shown to significantly reduce initial cracks, as evidenced by studies demonstrating that adding specific volume fractions of fibers can decrease the nominal total area of cracks by up to 96.8 %. It hinders the propagation of existing cracks, converting abrupt brittle fractures into progressive ductile failure mechanisms while substantially enhancing material toughness and long-term structural integrity.

Compared to natural fibers, synthetic fibers are non-renewable and non-biodegradable. Their waste disposal significantly contributes to environmental pollution, and the manufacturing process creates harsh conditions for factory workers. The production of synthetic fibers also leads to excessive wear and tear on manufacturing equipment and releases toxic gases and harmful chemicals into the atmosphere. Handling these materials can cause health issues, especially dermatitis. Despite their superior mechanical properties, synthetic fibers are ultimately unsustainable because of the various environmental sustainability problems they create [22]. Table 2 compared the advantages and disadvantages of synthetic fibers with natural fibers.

2.3 Comparison of physical and mechanical properties of fibers

The classification of reinforcement fibers plays a crucial role in determining the structural performance of fiber-

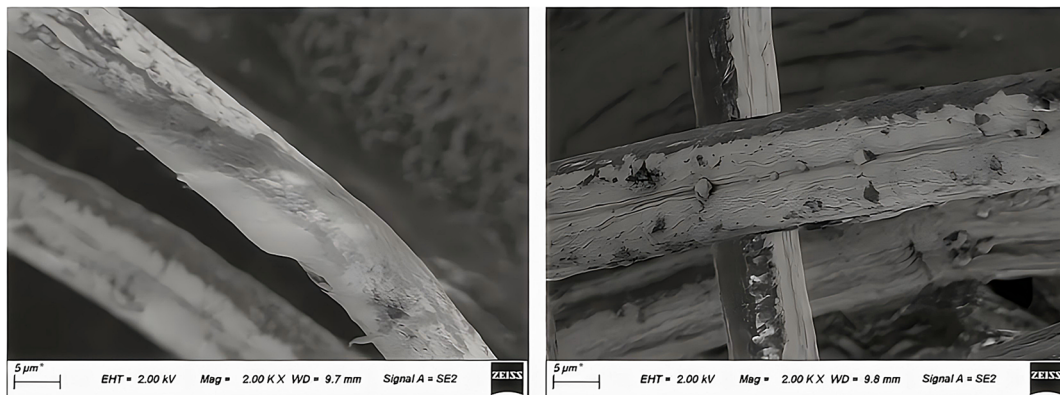


Figure 1: FE-SEM images of bamboo raw fibers treated with 3 % concentration of NaOH. Note: Figure 1 Cited from reference [29].

Table 2: Advantages and disadvantages of synthetic fibers [24].

Advantages	Disadvantages
Exceptional load-bearing capacity and structural integrity characteristics	Non-biodegradable
Moisture resistant	Not environment friendly
More durable than natural fibers	Poor insulation capacity
Avoid reliance on cultivated crops or livestock husbandry	Higher density compared to natural fiber, component weight increases
Shrinkless	
Rapid drying	
Superior deformation tolerance and abrasion resilience than plant-derived fibers	

reinforced concrete. Numerous investigations have been conducted on different fiber categories, examining their impact on key parameters such as compression resistance, bending capacity, fracture toughness under tension, stiffness characteristics, and response to dynamic loading conditions. Research outcomes demonstrate substantial variations in these mechanical attributes, as evidenced by the comparative data presented in Table 3.

The tensile strength of natural fibers ranges from 85 to 2000 MPa, while their modulus of elasticity varies between 2.7 and 85 GPa. Among various natural fibers, flax fiber stands out with the highest tensile strength and modulus of elasticity.

Lignin cellulose fibers have high moisture absorption levels, due to their hydrophilic properties and reduced mechanical performance, which ultimately compromise the structural integrity of the fibers. In order to ensure good mechanical behavior in fiber reinforced concrete, natural fibers should be limited to a length of less than 50 mm.

Another category of fiber is synthetic fibers, crafted from a diverse array of plastic materials, including steel, carbon, glass, aramid, polypropylene, polyvinyl alcohol, and polyethylene. Synthetic fibers exhibit outstanding tensile strength and elastic modulus compared to other natural fibers. Cellulose fiber boasts the highest tensile strength of 4137 MPa and modulus elasticity of 242 MPa, coupled with exceptional lightweight and specific stiffness, making it an ideal choice for composite materials.

Polyvinyl alcohol fiber (PVAf) demonstrates tensile properties between 1,200 and 1,600 MPa, whereas PPF exhibits strength values from 300 to 560 MPa alongside a density measurement of 0.91 g/cm^3 . When considering material characteristics, PPF displays reduced density and elastic modulus relative to alternative fiber types. Conversely, SF manifests tensile capabilities spanning 800–

1,345 MPa. The incorporation of PPF within concrete mixtures fails to deliver equivalent advancements across multiple performance metrics including structural rigidity, material density, and mechanical behavior. Primary motivations for utilizing polypropylene in concrete formulations center on improving flame retardant qualities. Among reinforcement options, Steel fibers dominate concrete applications due to their exceptional tensile resistance and elastic modulus properties. These metallic additives substantially boost multiple concrete attributes: resistance to compressive forces, residual tensile capacity after fracture, energy dissipation potential, and bending performance [33].

Glass fiber, which come in a wide range of diameters and lengths, have high rupture strength and are lightweight. glass fiber presents a superior substitute for organic textiles owing to its outstanding structural characteristics, including impressive fracture resilience, oxidation stability, superior mass efficiency, rigidity, minimal expansion properties, economic viability, and notable thermal endurance when employed as reinforcement in composite matrices. Nevertheless, glass fiber demonstrates suboptimal bonding at interfaces due to limited molecular attraction and inferior electrical/thermal transmission capabilities.

To fully leverage the benefits of synthetic and natural fiber-reinforced materials, it is advisable to utilize fiber-reinforced hybrid composites [34–38]. For instance, incorporating jute fibers into glass fiber-reinforced polymer composites enhances both modulus and tensile strength, as evidenced by recent studies.

Hybrid-reinforced concrete, which incorporates both natural and synthetic fibers, exhibits enhanced strength and durability relative to composites reinforced with only one type of fiber. For instance, flax fiber-reinforced polymer composites display superior flexural strength but lower tensile strength.

Carbon fiber can not only improve the mechanical properties of cementitious materials, but also impart additional functionalities such as electrothermal, piezoresistive, and electromagnetic shielding properties [39].

The good mechanical properties of composite materials are of great value in the aviation field. For example, aramid fiber has a high strength and modulus, good impact resistance and chemical stability, and the working temperature can be between 160 and 400 °C, which meets the requirements of aircraft engine cold end structure material working temperature > 300 °C [40].

Ultra-high molecular weight polyethylene fibers demonstrate exceptional mechanical properties, with a theoretical crystalline strength modulus of 31 GPa and a crystalline modulus of 316 GPa. Their density ranges from 0.96 to 0.98 g/cm^3 , making them the most compact and

Table 3: Mechanical properties of commonly used natural fibers and synthetic fibers [30–32].

Fiber type	Fiber length (mm)	Fiber diameter (mm/ μ m)	Tensile strength (MPa)	Modulus of elasticity (GPa)	Density	Elongation (%)	Water absorption (%)
Polyvinyl alcohol	8, 12	38 μ m, 40 μ m, 100 μ m	1,200, 1,560, 1,600	41	1,300 kg/m ³	–	
Polypropylene	12, 13, 15, 19	22 μ m, 800 μ m, 0.3, 18 μ m, 0.095, 0.022	480, >400, 560, 360, 300–400	4.1, 5.5	0.91 g/cm ³	–	
Cellulose	2.1	0.018	760	–	–	–	
Carbon	10, 20	6.9, 7.2, 7 μ m	4,137, 586–675, 3,530	242, 57.9–59.9, 230	1.81 g/cm ³ , 1.76 g/cm ³ , 1.78 g/cm ³	1.5, 1.6	6.5
Basalt	12	12 μ m	4.5	93–110	2.65 g/cm ³	3.15	0.02
Glass	15, 24, 36	0.54, 15.7 μ m, 1.0	>1,700, 1,000–1,700, 1,400	72, 74	2,680 kg/m ³ , 2,600 kg/m ³	–	1.7
Coconut	20–30	100–410 μ m	120–200	19–16	–	–	
Steel	30, 35, 60	0.7, 0.9, 0.54, 0.55	800–1,000, 1,150, 1,345, 1,100	200	7,850 kg/m ³	6–12	
Boron	–	100–40 μ m	4,000	70–190	2.49–2.57 g/cm ³	–	
Aramid	–	5–15	2,800–4,100	70–190	1.4 g/cm ³	2–2.4	
Kevlar			3,000	60	1.44 g/cm ³	2.5–3.7	5–7
Jute		25–250 μ m	200–800	10–55	1.35–1.46 g/cm ³	1.4–3.1	12.6
Sisal		50–200 μ m	880	9–38	1.2–1.5 g/cm ³	2–25	
Hemp		25–600 μ m	310–900	17–70	1.4–1.5 g/cm ³	1.6–6	10
Flax		25 μ m	85–2,000	20–85	1.4–1.54 g/cm ³	1–10	10
Bamboo		88–125 μ m	140–230	10–30	0.6–1.0 g/cm ³	1.3–3.7	9.9
Kenaf		65–71 μ m	240–1,195	2.85–55	1.2–1.6 g/cm ³	1.6–6.9	–
Palm			309	2.7–11	0.7–1.0 g/cm ³	2.5–13	
Banana		100–250 μ m	350–910	8–33	1.3–1.35 g/cm ³	3–10	8.7
Pineapple		50 μ m	126–1,600	4.4–82	1.4–1.56 g/cm ³	0.8–3.2	13
Roselle			145–290	1.75–2.76	0.75–0.85 g/cm ³	2.2–8	

highest specific modulus and strength fiber material developed to date. Compared to aramid fibers, UHMWPE exhibits 40 % greater tensile strength and operates within an exceptionally wide temperature range of 150 to 100 °C. The total impact absorption energy of this material is 1.8 times that of carbon fiber, 2.6 times that of aramid fiber, and 3 times that of glass fiber, demonstrating its exceptional impact absorption capability. Leveraging these outstanding properties, ultra-high molecular weight polyethylene fibers have seen expanding applications in military industries [41].

3 Fiber applications

3.1 Natural fiber applications

Due to their natural fiber chemical and mechanical properties, cost-effectiveness, biodegradability, and environmental benefits, they exhibit significant potential for utilization in advanced composite systems and sustainable bioplastic formulations.

Natural plant-derived fibers like hemp, flax, jute, and kenaf are incorporated within synthetic polymer bases, including epoxy, polyester, or eco-friendly resin formulations, producing materials that combine lightness with structural integrity. These bio-based composite materials gain recognition for possessing favorable weight-to-strength ratios, minimal ecological footprint, and advantageous mass characteristics. For example, automotive manufacturers use composites made from hemp or flax in door panels, dashboards, and trunk liners to decrease vehicle weight and enhance fuel efficiency. The construction industry also utilizes these materials in insulation boards, wall panels, and furniture. Bamboo-reinforced structures are commonly used in urban areas of bamboo-rich countries like China. Some researchers have conducted performance evaluations on bamboo-reinforced concrete beams. El-Messiry and colleagues (2017) presented novel research outcomes, showing that adding flax pultruded reinforcement substantially boosted impact performance, achieving an impressive 5–10 fold enhancement compared to standard cement mixtures [42].

In bioplastics, natural fibers are integrated into polymers derived from renewable sources such as polylactic

acid (PLA), starch, or cellulose – to enhance mechanical properties and reduce reliance on fossil fuels. For example, flax or wood fibers blended with PLA create rigid, heat-resistant bioplastics for packaging, disposable cutlery, and 3D printing filaments. Similarly, coconut coir or rice husk fibers reinforce starch-based bioplastics, improving tensile strength and water resistance for agricultural mulch films or single-use products. These hybrid materials retain the biodegradability and low-carbon advantages of bioplastics while addressing brittleness and thermal limitations. Medical applications are also emerging, with silk or chitosan fibers reinforcing biocompatible plastics for sutures and implants.

3.2 Synthetic fiber applications

Synthetic fibers are commonly utilized to enhance the structural components of buildings, including beams, slabs, and columns, by incorporating them as sheets, laminates, and bars [43].

Over the past two decades, the Netherlands has seen the construction of numerous synthetic fiber-reinforced polymer (SFRP) bridges. In early 2002, Australia's first composite fiber bridge demonstrated the viability of SFRP composites as structural materials for bridges. A notable instance of fiber-reinforced polymer (FRP) application involves a structural steel bridge in Guthrie County, Iowa, where engineers implemented carbon fiber-reinforced polymer (CFRP) tendons through post-tensioning techniques [44].

CFRP have widely used in many fields such as aerospace, wind power generation, automotive, and civil engineering. However, the associated waste is also on the rise. For the aerospace sector, the total amount of 500,000 tons of cumulative CFRP waste is expected to be generated by 2050 [45]. Proper treatment methods can reduce land and air pollution, which is crucial for sustainable environmental development. Currently, the most economical method is recycling, recycled CFRP, as a necessary strategy for sustainable material management, is crucial from the perspectives of technology, economy and ecological protection [46]. And a quantitative environmental sustainability assessment model combined with life cycled assessment or economic cost analysis should be developed to evaluate the environmental impact of CFRP. Flexural, tensile and splitting tensile strength of recycled CFRP-reinforced mortar increased 20.3 %, 35.5 % and 44.6 % than mortar without fiber, respectively. The recycled CFRP-reinforced mortar achieved a 19 % higher integrated benefit index than mortar without fiber. Microencapsulated phase change materials (mPCMs) are energy-saving and environmentally friendly

green materials that can release and absorb latent heat. Incorporating mPCMs into cement concrete for pavements can improve the antifreeze performance of concrete and delay icing. Liu et al. (2024) investigates the influence of different mPCM dosages on the working performance, mechanical properties, frost resistance, and pore structure of concrete. Results showed that the mPCMs can effectively regulate the temperature within and on the surface of the concrete [46].

Adding polypropylene fibers to recycled concrete can compensate for its poor toughness, but this approach reduces the material's mechanical properties. Incorporating nanomaterials such as nano-calcium carbonate, nano-silica, and nano-alumina enhances hydration rates while improving flexural and compressive strength. Researchers worldwide have extensively studied composite-reinforced recycled concrete, determining optimal blending ratios through microscopic morphology analysis and hydration studies. Wei (2022), building on Krstulovic-Dabic's kinetic model of cement-based materials' hydration processes, developed the following kinetic equations [47]:

$$da/dt = F_1(\alpha) = K_1' n (1 - \alpha) [-\ln(1 - \alpha)]^{(n-1)/n} \quad (1)$$

$$da/dt = F_2(\alpha) = K_2' 3 (1 - \alpha)^{2/3} \quad (2)$$

$$da/dt = F_3(\alpha) = K_3' 3 (1 - \alpha)^{2/3} / [2 - 2(1 - \alpha)^{1/3}] \quad (3)$$

where: α represents the degree of cement hydration, K_1' (K_1') is the reaction rate constant in the three processes, and t represents the end time of cement induction period.

In the context of earthquake-resistant upgrades for concrete buildings, SFRP composites serve as eco-friendly solutions for improving structural resilience. These composite materials integrate robust fibers within a polymer base, effectively redistributing substantial forces while protecting the reinforcement from harsh environmental exposure. Extensive research has documented the effectiveness of SFRP-strengthened concrete columns during seismic events. Ghatte's findings demonstrated that both CFRP wrapping and supplementary steel confinement offer superior advantages compared to conventional reinforcement techniques, particularly regarding deformation tolerance and energy absorption capabilities. Chandrathilaka et al. 2021 examined the performance and behavior of polyethylene fiber-reinforced concrete (PEFRC) under projectile impact, with the aim of utilizing PEFRC in structures resistant to impact and blasts. In contrast to standard concrete mixtures, PEFRC demonstrated superior resistance to shattering along with decreased occurrences of surface spalling, material fragmentation, and overall damage area. Nevertheless, experimental observations revealed

comparable penetration depths between PEFRC and conventional concrete formulations [48].

Although synthetic fiber concrete improves the bearing strength of buildings, the bearing performance is also related to the design form and construction technology of components, CFPR filling with a bio-inspired hexagonal-quadrilateral lattice structure (HQLS) can solve the limitation of premature buckling and insufficient energy dissipation, this kind of lightweight composite structures are increasingly utilised in rail vehicle crashworthy systems [49]. Wang et al. (2025) studied load-bearing capacity of grout-lifted compressible concrete-filled steel tube composite column, the ultimate bearing capacity calculation formula for compressible columns was also derived [40].

4 Conclusion and future prospective

Studies have explored the properties of natural and synthetic fibers, as well as advancements in fiber and fiber hybrid polymer composites, highlighting their potential to enhance mechanical performance.

Natural FRP has emerged as a sustainable alternative to traditional composites, contributing to a circular economy. When natural fibers are incorporated in the right proportions, they can enhance tensile strength, flexural strength, and impact performance. However, the effects of different fiber particles on the mechanical properties of FRP are still debated, with mixed results regarding tensile and impact performance. Further research is needed to clarify the link between particle size and mechanical reinforcement potential. Additionally, comparing the reinforcement potential of various composites to traditional synthetics provides valuable insights. Efforts are ongoing to improve the mechanical performance, durability, and cost efficiency of these materials.

The improvement of fiber synthesis technology plays an important role in improving the mechanical properties of materials. For example, from the perspective of rheology, printing materials become the direction of the optimisation performance of models [50]. The combination of 3D printing polymer materials and microwave heating technology for post-processing requires further discussion on its regulatory mechanism and microwave coupling mechanism, so that it can be applied in aerospace and other important fields.

This study establishes a conceptual framework for experts and analysts working on eco-friendly material innovation. The examination of key discoveries related to plant-derived fibers and their combination with synthetic polymers

will support scientists in progressing bio-based composite materials, helping them overcome obstacles encountered by contemporary manufacturing sectors. There remain numerous possibilities for exploring various kinds of natural fibers. Additionally, applying this understanding of material science is expected to open new paths to sustainability.

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